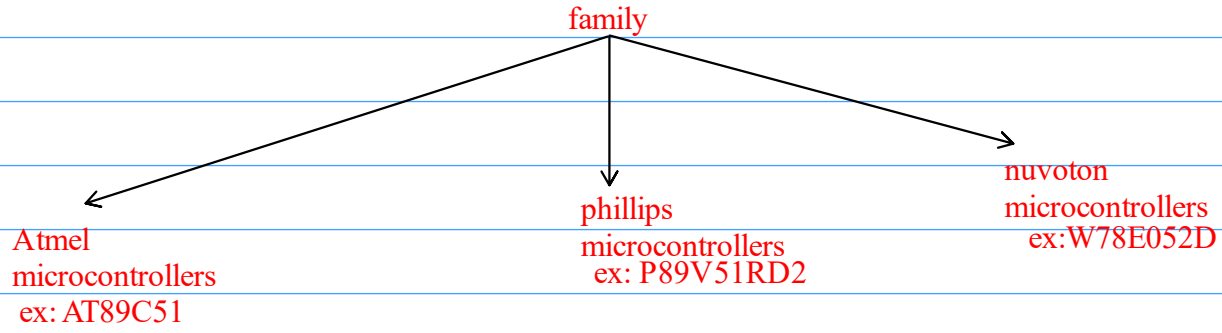
8051

8051 is a family of INTEL core 8-bit microcontrollers.
For this INTEL core, peripherals are integrated by different manufacturers like ATMEL,PHILIPS etc.



AT89C51 - basic 8051 microcontroller
20pins, 28 pins, 40 pins, 42 pins etc.

4 ports are there, Each port having 8 pins
4 x 8 pins = 32 port pins are available

P0, P2, P3 are having the dual functionalites. P1 is giving support for only GPIO operations.

by default, all port pins are giving support for GPIO operations.

40 pin DIP(40(physical)=32_I/O+8,64(logical)=(24*2)+16)

Physically:
P0, P1, P2, P3 - 4X8=32 PINS
VCC, GND, RST, XTAL1, XTAL2, EA/VPP, ALE/PROG, PSEN - 8 PINS
32+8 = 40
Logically:

3 ports are having the dual functionalities - 3portsx8pinsx2funcs = 48
port1 - 8
remaining - 8 (48+8+8=64)

AT89C51 - 4KBytes of on chip ROM (program memory)
- 128 bytes of on chip RAM (data memory)
- 128 bytes address space reserved for SFRs, available ONLY 21 bytes/21 sfr's

AT89S52 - 8KBytes of on chip ROM (code memory)
- 256 bytes of on chip RAM (data memory)
- 128 bytes address space reserved for SFRs, available more than 21 sfr's

Two 16 bit timers/counters peripheral (TIMER0, TIMER1)
TIMER0 - TH0, TL0 (8-BIT REGISTERS ARE AVAILABLE IN CASCADED)
TIMER1 - TH1, TL1 (8-BIT REGISTERS ARE AVAILABLE IN CASCADED)

8031 - ROMLESS controller

PORT0 - P0.0 (AD0) to P0.7 (AD7)
- Address and data signals are multiplexed (used for interfacing external memory)

PORT1 - P1.0 to P1.7

PORT2 - P2.0 (A8) to P2.7 (A15)
- higher byte address lines are used for connecting the external memory

PORT3 - dual functions

P3.0 - RXD (UART operations - used to receive data from external UART based device)
P3.1 - TXD (UART operations - used for sending data to external UART based device)
P3.2 - $\overline{INT0}$ (external interrupts)
P3.3 - $\overline{INT1}$ (external interrupts)
P3.4 - T0 (Timer/counter operations - must use this pin for counter0 operation)
P3.5 - T1 (Timer/counter operations - must use this pin for counter1 operation)
P3.6 - WR (control signal for writing data to the external memory)
P3.7 - RD (control signal for reading data from external memory)

Interrupt Control logic which supports 6 sources,5 vectored interrupts
reset, external interrupt0, external interrupt1, timer0 interrupt, timer1 interrupt & serial interrupt

Can support up to max 64k of ROM (hence 60k ext)
EA(bar)/VPP = 1, onchip 4K + offchip 60K
EA(bar)/VPP = 0, total 64Kbytes it will access from externally by neglecting onchip memory

EA - External Access
why maximum 64Kbytes?
total address lines are 16, i.e. maximum you can connect 2pow16 - 65536 - 64Kbytes

Mnemonic is used for code memory (internal/external) access is MOVC

Can support up to addition max 64k ex-RAM
Mnemonic is used for accessing the internal data memory is MOV & other Mnemonics (ADD, SUBB etc)
Mnemonic is used for accessing the external data memory is MOVX